CSE590 · MIDTERM PROGRESS

When and why does ApET work?

Understanding reconstruction-based visual token compression in LLaVA-1.5-7B

APET IN ONE LINE

Without attention, keep visual tokens that are hard to reconstruct linearly; average the rest into their most similar kept token.

SIMPLE, AND STRONG

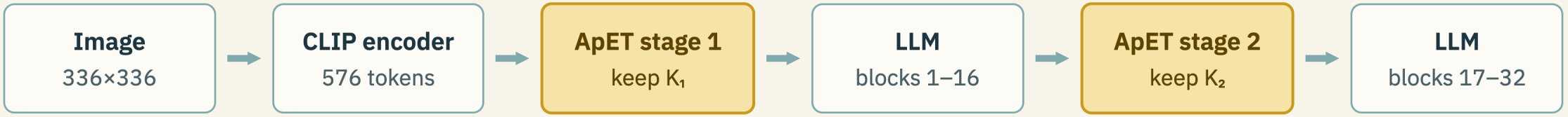
With 64 of 576 tokens, LLaVA-1.5 keeps 95.2% of its performance (VisionZip: 92.7%) and runs 1.46× faster.

OUR QUESTION

How can a linear, geometric criterion work inside a nonlinear VLM, and when does it fail?

Peng Chen · SBU ID 117702477 · October 2026 · Middle-column numbers from Ma et al., CVPR 2026

Method and three research questions



Setup: LLaVA-1.5-7B · native ApET code · Kaggle 2×T4, locked environment · GQA 800, MMStar 1,500, TextVQA 100 · avg. 64 or 192 tokens

RQ1 · Does E track answer relevance?

30-question diagnostic set. Information Horizon (IH) test: keep only one token, measure its effect on the answer, correlate with E.

RQ2 · Selection or merging?

Planned at matched budgets: E vs. random selection; merge vs. drop vs. random or protected merge targets.

Pilot: random sometimes beat ApET, so we first ran ApET vs. random at scale.

RQ3 · A lightweight improvement?

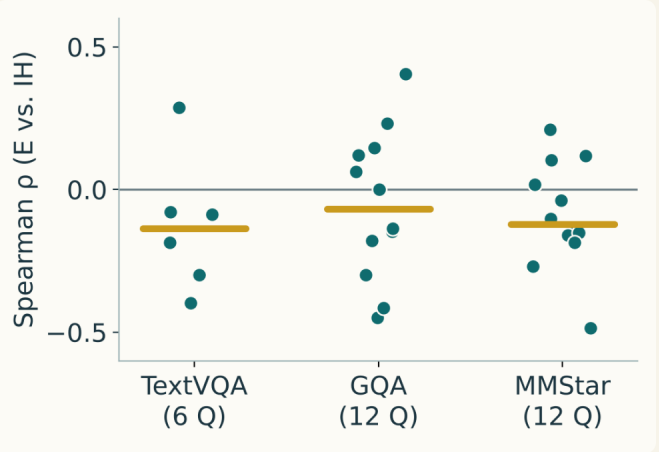
Only if the analysis supports it, e.g., an adaptive token budget. Exploratory.

E = reconstruction error from 10 basis tokens; merging averages each dropped token into its most similar kept token. · 2 / 5

ApET works on average, but not answer by answer

① E does not predict which tokens matter

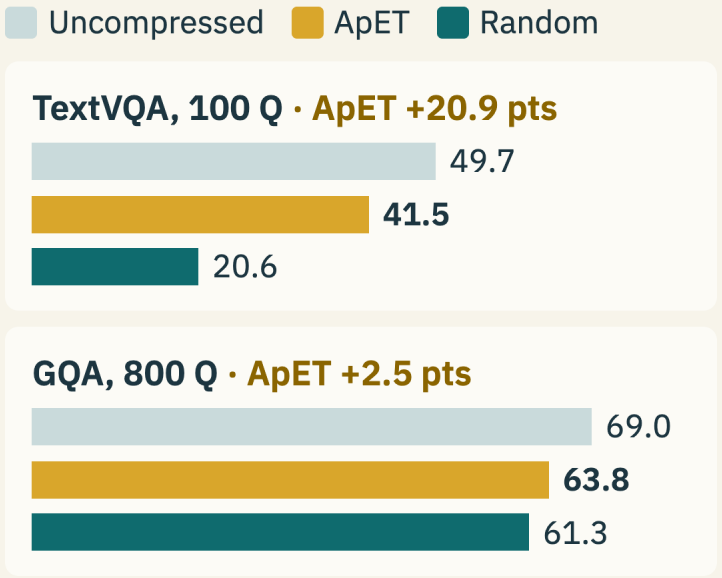
IH test: keep only one of 576 visual tokens and measure how much it raises the answer's probability. 30 questions, 17,280 token tests.



Medians −0.07 to −0.14; only 11 of 30 questions positive.

② Yet ApET beats random pruning, task-dependently

Accuracy (%) at the same budget (avg. 64 tokens); random = mean of 3 seeds.



③ Similar accuracy hides answer turnover

ApET versus the uncompressed model, at avg. 192 tokens:

GQA · 800 questions

34 lost · 26 gained

Accuracy 69.0% → 68.0%

MMStar · 1,500 questions

36 lost · 40 gained

Accuracy 34.7% → 35.0%

Lost = correct → wrong. **Gained** = wrong → correct.

IH: Information-Horizon-style retain-one test [Wang et al. 2026] · ① uses 30 hand-picked diagnostic questions · 3 / 5

Roadblocks and fixes

Independent issues, each with its own fix and status

Roadblock	Problem	Attempted fix	Current status
Integration	Kaggle package updates broke ApET/LLaVA compatibility.	Locked environment with recorded fixes.	Green: resolved
Run-to-run variation	Fresh compressed runs changed a few answers; replays from saved features were identical.	Diagnostic traced it to the merge step (likely order-dependent FP16 sums); keep first runs, repeat critical cases.	Yellow: managed, cause not isolated
Selection bias	The 30-question set was picked from ApET's own outcomes, so it favors other methods.	Use it only for RQ1 and case studies; compare components on unselected subsets.	Yellow: handled by design
Planning risk	Component attribution may stay unclear; no RQ3 candidate yet.	Report a well-controlled negative result if needed.	Yellow: open

Merging uses CUDA scatter_add_; GPU additions can run in a different order each time, and FP16 addition is not exactly associative.

What's next

01 · RQ2 selection

Does linear reconstruction pick what the model uses?

Vary only selection, merging fixed. For low-E tokens: kept, merged, dropped, or replaced by their linear reconstruction.

02 · RQ2 merging

Should we merge, and how?

Merge vs. drop vs. random or protected targets; measure how averaged tokens differ from the originals.

03 · RQ3, exploratory

A lightweight improvement?

Only if the analysis supports it, e.g., a question-aware budget. No guarantee.

SUCCESS CRITERION

One controlled comparison isolating a single ApET component on ≥100 unselected questions: matched budgets, 3 seeds, paired 95% intervals. Negative results count.